<table>
<tr><td>Course Title</td><td colspan="3">Computational and Numerical Methods</td></tr>
<tr><td>Course Code</td><td>CS 374</td><td>Credit Structure</td><td>3-0-3 -4.5 (L-T-P-C)</td></tr>
<tr><td>Category</td><td>Core</td><td>Semester</td><td>Autumn Semester (AY 25-26)</td></tr>
<tr><td>Program</td><td colspan="3">B.Tech-ICT-CS</td></tr>
<tr><td>Prerequisites</td><td colspan="3">NA</td></tr>
<tr><td>Course Objectives/ Brief Course Description</td><td colspan="3">There are several problems in applications that can’t be solved analytically. For that reason people search for approximate solutions. The aim of this course is to make the students learn some numerical techniques to find the approximate solutions of a variety of problems such as numerical solutions of algebraic and transcendental equations, numerical solutions to systems of algebraic equations, interpolation, numerical differentiation, numerical integration, numerical solutions of differential equations. For hands-on experience in problem solving, students will be writing in some of the programming language of their choice.</td></tr>
<tr><td>Evaluation/ Grading Policy</td><td colspan="3">Labwork: 30%
Insem – 1: 20%
Insem – 2: 20%
Final Exam: 30%</td></tr>
<tr><td>Course Materials/ References</td><td colspan="3">Lecture notes.
Lecture notes/ class notes will be posted and made available for students after the lecture.
Labs
In labs the students will do hands on to implement the numerical methods using a programming language. Instructor and TA will help the students in labs
Books
1. Elementary Numerical Analysis, K. Atkinson and W. Han, Wiley & Sons.
2. Numerical Analysis, Mathematics of Scientific Computing, D. Kincaid, W.Cheney, American Mathematical Society.
3. An Introdution to Numerical Methods and Analysis, James F. Epperson, Wiley and Sons.
4. Numerical Method for Scientific and Engineering Computation, M. K. Jain, S. R. K. Iyengar, and R. K. Jain, New Age International Publishers.
5. Computer Oriented Numerical Methods, V. Rajaraman, Prentice Hall of India</td></tr>
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Detailed Course Content	***Computer Arithmetic***- Floating Point Numbers, Errors ***Root finding-*** Numerical solutions to Algebraic and Transcendental equations, Bisection method, Secant method and Newton Raphson method, Fixed point iteration method. ***Interpolation-*** Lagrange interpolation, Divided difference Interpolation, Newton's forward and backward interpolation, Piecewise interpolation and Spline Interpolation. ***Numerical Differentiation and Integration-*** Numerical differentiation using Lagrange interpolation, using difference operators, and using method of undetermined coefficients, Numerical Integration, Trapezoid rule, Simpsons rule and Gaussian quadrature rule. ***Solution of Systems of Linear Equations-*** Numerical Solutions to System of Algebraic equations, Gaussian elimination method, Gauss Jordan method, Gauss-Seidel iteration method, Jacobi iteration method, Numerical solution to nonlinear equations, Newton's method using Jacobian *Ordinary Differential Equations*- Numerical Solution of Differential equation, Euler's method, Midpoint rule, Second order Runge-Kutta method, fourth order Runge-Kutta method, Application to boundary value problem

Course Outcome: The students after completing the course will get

CO1: A basic overview of Numerical methods.

CO2: Students will be able to understand the well-known numerical methods for solving problems in science and engineering.

CO3: After the completion of the course the students will be able to understand which numerical methods should be applied to a particular problem.

POs-COs Matrix:

P1	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**P8**	**P9**	**P10**	**P11**	**P12**
x	x	x	x	x							x

APPENDIX: Detailed Course Content (Session-wise/ Module-wise)

Tentative Lecture Schedule

Sl. No.	Description	No. of Lectures
1	**Introduction to Numerical Analysis, Computer Arithmetic**	**3**
2	**Root Finding:** Bisection Method, Secant Method, Newton-Raphson Method, Fixed Point Iteration Method	**7**
3	**Interpolation:** Lagrange Interpolation, Divided Difference Interpolation, Newton's forward and backward Interpolation, Piecewise Interpolation, Spline Interpolation	**7**
4	**Numerical Integration and Differentiation:** Trapezoid Rule, Simpson's Rule, Gaussian Quadrature Rule, Differentiation Using Interpolation, Differentiation using Method of undetermined coefficients	**8**
5	**Solution of Systems of Linear Equations:** Gauss Elimination Method, Gauss Jordan Method, Gauss-Seidel Method, Jacobi Iteration Method, Newtons Method for solving nonlinear systems	**6**
6	**Numerical solutions to Ordinary Differential Equations:** Euler's Method, Midpoint Rule, Second Order and fourth order Runge-kutta Method, Boundary Value Problems	**8**

Week	Topics
1	Introduction and Computer Arithmetic
2	Bisection Method, Secant Method
3	Newton-Raphson Method, Fixed Point Iteration Method
4	Lagrange Interpolation, Divided Difference Interpolation
5	Newton's forward and backward Interpolation, Piecewise Interpolation

6	Spline Interpolation, Trapezoid Rule,
7	Simpson’s Rule, Gaussian Quadrature Rule
8	Differentiation Using Interpolation, Differentiation using Method of undetermined coefficients
9	Gauss Elimination Method, Gauss Jordan Method
10	Gauss-Seidel Method, Jacobi Iteration Method
11	Newtons Method for solving nonlinear systems
12	Euler’s Method, Midpoint Rule
13	Second Order and fourth order Runge-kutta Method,
14	Boundary Value Problems